PART 02 · CHAPTER 09

Breakouts, pullbacks & failed moves

Define the boundary and the observation time before evaluating a crossing.

Answer edition: includes continuation, worked solutions and the knowledge-check answer.

beginner About 8 min of prose · practice takes additional time AI editorial review 2026-10-02 · v0.4

▼ Review scope

Scoped chapter editorial acceptance: prose, material source claims, tasks, numerical reproduction and compact/wide figures; release QA and human learning are separate.

Recorded by /root. The review covers editorial evidence; it does not establish learning by human readers.

> THE IDEA TO KEEP
>
> A breakout is a predefined crossing, not a promise. Confirmation and retests trade additional information for changed timing and execution.

By the end of this chapter

- Determine when a fixed closing-breakout rule becomes observable.
- Separate signal recognition, an eligible entry, and a later failure.
- Reproduce the quantity, costs, and R result of a fully stated paper plan.
- Retain successful, failed, skipped, and ambiguous continuations in a review.

Builds on: The language of the market · Volume: participation behind price · Timeframes, sessions & chart scales · Support, resistance & price areas · Classical chart patterns

A crossing needs a boundary and a clock

A breakout is a move beyond a previously identified price boundary or area. The word by itself is incomplete: beyond which edge, by how much, using what price field, and at what observation time? An intraday high above resistance and a completed close above it are different conditions. Both are legitimate observations, but they define different events. A downward crossing is often called a breakdown.

Imagine Harbor has a resistance area from $49.70 to $50.00. A trade at $50.01 crosses the upper edge, yet a rule requiring a close more than 1% above that edge needs a close above $50.50. The small buffer is an illustrative convention, not an optimized or universally effective threshold. Fix it beforehand. A level selected after seeing the crossing cannot fairly demonstrate that you recognized a breakout in advance.

The exact price field changes the result

Suppose a fictional daily bar opens at $49.40, reaches $50.80, falls to $49.10, and closes at $49.70. It crossed $50 intraday but did not finish above it. An intraday-break rule triggered; a close-above-$50 rule did not; a close-above-$50.50 rule also did not. Describing the move as false without naming the rule hides these differences. The label should refer to the event you actually defined.

If you require the completed close, that information becomes available at the end of the session. A backtest must then use a feasible subsequent execution convention, or an explicitly justified closing-auction model with its actual order and information deadlines. It cannot simply observe the finalized closing signal and assume a guaranteed trade at the same price. The decision clock is part of the method. Treating timing as an afterthought can make a historical breakout system appear better than it could have been.

First pass: write the Harbor instruction at the session-9 close

The first two sections distinguish price fields and observation times. Pause the main fictional Harbor case at the completed session-9 close. The resistance area $49.70–$50.00 and rules below were supplied before this cutoff; they are not selected from the future path. All nine closes and the complete session-9 candle are shown. The next session has not opened.

Apply the closing signal rule. Record what is observable now and what cannot yet be recorded about entry eligibility, a filled position or a later failure. Write the exact next-opening gate, including equality at both edges, the deadline and the behavior after a rejected open.

Calculate the maximum-entry cash reservation including the entry fee, the maximum allowed entry's price-risk 1R and its ordinary cost-inclusive stop loss. Give the formula that freezes 1R if the eligible opening entry is lower. State which later bar fields and modeled execution assumptions would be needed to calculate a realized result.

Save the cutoff note before revealing the full Harbor path and the separate comparison branches. The volume-filter and retest examples later in the chapter are different rules; neither is an extra gate for this main plan. The constructed paths teach accounting and chronology, not return probabilities or a recommended allocation.

Completed Harbor closes available through session 9 (fictional USD)

Session	Close ($)
1	48
2	49
3	49.8
4	49.2
5	48.7
6	49.6
7	49.9
8	49.4
9	50.8

Session-9 completed candle only (fictional USD)

Session	Open ($)	High ($)	Low ($)	Close ($)
9	49.4	51	49.2	50.8

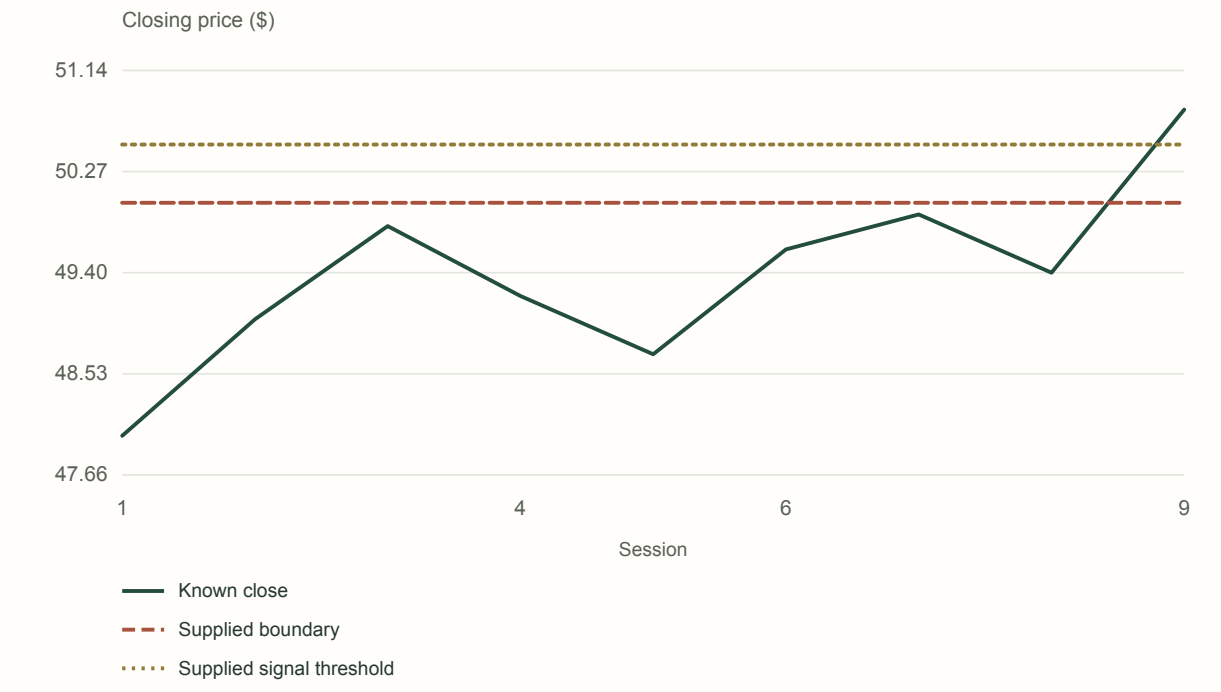
Harbor rules supplied before the continuation

Field	Frozen main-plan instruction
Signal	Completed regular-session close strictly above $50.50; no volume gate.
Opening entry	Only the next session's open: accept strictly above $48.80 and at or below $50.80. Model 50 shares at that opening price; otherwise skip.

Field	Frozen main-plan instruction
Deadline / rejection	Opening-only decision. A rejected open expires the candidate; a later price return does not permit an entry.
Account assumptions	Sufficient available cash; one position; no borrowing. Reserve maximum entry notional plus the $2.50 entry fee.
Standing exits after entry	Sell stop $48.80; sell-limit target $54.80. Exact trigger/target fills are teaching assumptions when their modeled conditions are met.
Gap / ambiguous bar	A later opening gap through the stop exits at the modeled opening price. Without an opening event deciding the first exit, a bar spanning both exit levels is sequence-ambiguous; do not invent an order.
Time exit	If still held, model exit at the third post-signal session's close. Standing exits apply first.
Chart failure	Any of the next three completed closes strictly below $50.00. Equality is not failure; this chart label is separate from the trade outcome.
Fees / R	Fixed $5 round trip: $2.50 on entry and $2.50 on exit. Freeze price-risk 1R after entry as 50 × (filled entry – $48.80); fees are separate.

FICTIONAL TEACHING ILLUSTRATION

Harbor observations available at the session-9 close



Fictional completed closes through session 9. The $50 boundary and strict $50.50 signal threshold were supplied in advance. No next-session candle, gap or later close appears.

▼ View the illustration's data

Harbor observations available at the session-9 close

Session	Known close	Supplied boundary	Supplied signal threshold
1	48	50	50.5
2	49	50	50.5
3	49.8	50	50.5
4	49.2	50	50.5
5	48.7	50	50.5
6	49.6	50	50.5
7	49.9	50	50.5
8	49.4	50	50.5
9	50.8	50	50.5

Compare the continuation with your frozen note. This reference includes outcome headings, complete ledgers and solutions. In the learner print edition it starts on a new page after the cutoff packet.

Confirmation filters add requirements, not certainty

Some rules require a buffer, repeated closes, increased volume, or several conditions together. These filters select a different set of observations and often delay recognition. They may remove some failed moves and exclude some successful ones. Whether the tradeoff is useful depends on measurement across the relevant universe and period. Calling a condition confirmation does not confer statistical independence or proven effectiveness.

Consider an illustrative close above $50 with volume at least 1.5 times the prior twenty-session mean. A $50.80 close on 1.6 times volume satisfies it. A $52 close on 1.2 times volume does not, even if price keeps rising later. That exclusion is part of the rule rather than an error to correct with hindsight. If you relax the volume requirement only on the examples that subsequently succeed, you are changing the method after learning the outcome.

Retests are optional paths

After an upward crossing, price may revisit the former boundary; after a downward crossing, it may revisit from below. Retest, pullback, and throwback terminology varies, so this book describes the direction explicitly. A return toward the area is an observation. A supportive reaction or renewed decline requires another defined condition. Price may also continue without returning at all.

If Harbor closes at $51 beyond a $50 boundary, a retest rule might require a later low inside $49.80–$50.20 followed by a close above $50.50 within three sessions. This original convention makes the hypothesis testable. A low of $50.30 misses the specified visit, and a low of $49 followed by a $49.50 close does not meet the supportive-close condition. Waiting can improve the prospective entry price in some paths, miss others, or encounter an actual failure. Record all three possibilities.

Execution can change the economics

Suppose the illustrative signal close is $50.80, a planned protective trigger is $48.80, and a geometric reference is $54.80. At the signal price, nominal downside distance is $2 and upside distance is $4. The next open is $52.20. Entering then changes those distances to $3.40 and

$2.60, respectively. Their ratio changes from 2.0 to about 0.76. Neither calculation is an expectancy because it supplies no probabilities or execution guarantees.

An overnight gap can also pass an intended order level entirely. A market order seeks execution but does not guarantee the displayed price; a limit sets a price constraint but may not fill; a stop-market order activates a market order when its trigger is reached. These distinctions matter when a chart makes an entry look exact. A valid signal can lead to a poor executable opportunity under the chosen plan, and declining that opportunity is consistent with having recognized the chart event accurately.

Define failure without selecting the answer

A false-break definition needs a time window and a return condition. One illustrative rule calls an upward closing breakout failed if any of the next three completed closes is below the original $50 boundary. Another requires price to hit a protective trigger. They are different outcome measures. Neither can label the event failed at the instant of the initial crossing; the later observations have not happened yet.

A failed upward move does not automatically prove a profitable short trade. Reversing direction introduces a new entry, costs, risk, and a hypothesis that can fail independently. Price may oscillate across the area repeatedly. Study the full sequence instead of drawing arrows only on the profitable sections. Breakout analysis becomes useful when it forces you to specify boundaries, timing, execution, and update rules. Its value should be evaluated through that complete process, not through a collection of memorable escape-from-the-range pictures.

Freeze a small plan before revealing the continuation

Return to the main Harbor chart. Its session-9 close is $50.80 and session-10 close is $49.40. Before session 9, the resistance area is already fixed at $49.70–$50.00. For this exercise, use a completed regular-session close strictly above $50.50 as the signal. Volume is not an additional condition in this main rule; the earlier volume-filter example is a separate comparison. Declare failure if any of the next three completed closes is strictly below $50.00. Equality at either threshold does not qualify. These arbitrary rules are supplied so two readers can classify the same sequence without supplying missing decisions.

Freeze the following paper execution model after the session-9 close. Use only the next session’s opening observation. If that open is above $50.80, skip the candidate entirely, even if price later returns. Otherwise model 50 shares filled at the open, provided it is above the $48.80 protective trigger. An open at or below the protective trigger is also a skip. Model a sell stop at $48.80 and a sell limit target at $54.80; if neither exits the position by the third post-

signal session, model an exit at that session's close. A later opening gap through the stop uses the opening price, not $48.80. If an ordinary OHLC bar spans both exits without an opening gap deciding the first event, mark the result sequence-ambiguous rather than inventing an order. All fills here are explicit teaching assumptions, not deductions from a real order book.

At the maximum allowed entry of $50.80, 50 shares have $2,540 entry notional before costs and $100 planned price risk: 50 × ($50.80 − $48.80). This paper plan assumes sufficient cash, one position, no borrowing, and a fixed $5 round-trip expense, charged $2.50 on entry and $2.50 on exit. Reserve $2,542.50 before entry to cover the maximum entry debit. Freeze 1R after the modeled fill as 50 × (entry price − $48.80); an eligible lower entry has a smaller price-risk denominator. At $50.80, the ordinary cost-inclusive modeled loss is $105, or −1.05R net, and the exact target exit is +$195, or +1.95R net. Costs are separate from the price-risk denominator; a $100 all-in loss budget would require a smaller quantity at this entry. A gap can exceed either budget.

Now reveal the main continuation: session 10 opens at $52.20, reaches $52.60, falls to $49.10, and closes at $49.40. At the open the plan records a skip; at the close the chart event meets the failure definition. No trade occurred, so the journal records no realized trade P&L or R outcome. The hypothetical distances from a chased $52.20 entry remain useful for comparison, but do not become this plan's account result. Observation, eligibility, and outcome each have their own timestamp.

One frozen rule, three supplied continuations

Branch	Information after the $50.80 signal	Decision and modeled result
Main gap/skip	Next O52.20/H52.60/L49.10/C49.40	Skip at open; closing failure later; no trade P&L
Target comparison	Next O50.80/H55.00/L50.60/C54.90	Assumed entry 50.80; target 54.80; +$195 net = +1.95R
Stop comparison	Next O50.80/H51.00/L48.60/C49.40	Assumed entry 50.80; stop 48.80; −$105 net = −1.05R

FICTIONAL TEACHING ILLUSTRATION

A closing breakout followed by a return below



Main Harbor closes use the same session-9 $50.80 signal and session-10 $49.40 failure as the worked record. The fixed $50 boundary and stricter $50.50 signal threshold are distinct. The supplementary candles expose the $52.20 opening gap that this close-only line omits.

▼ View the illustration's data

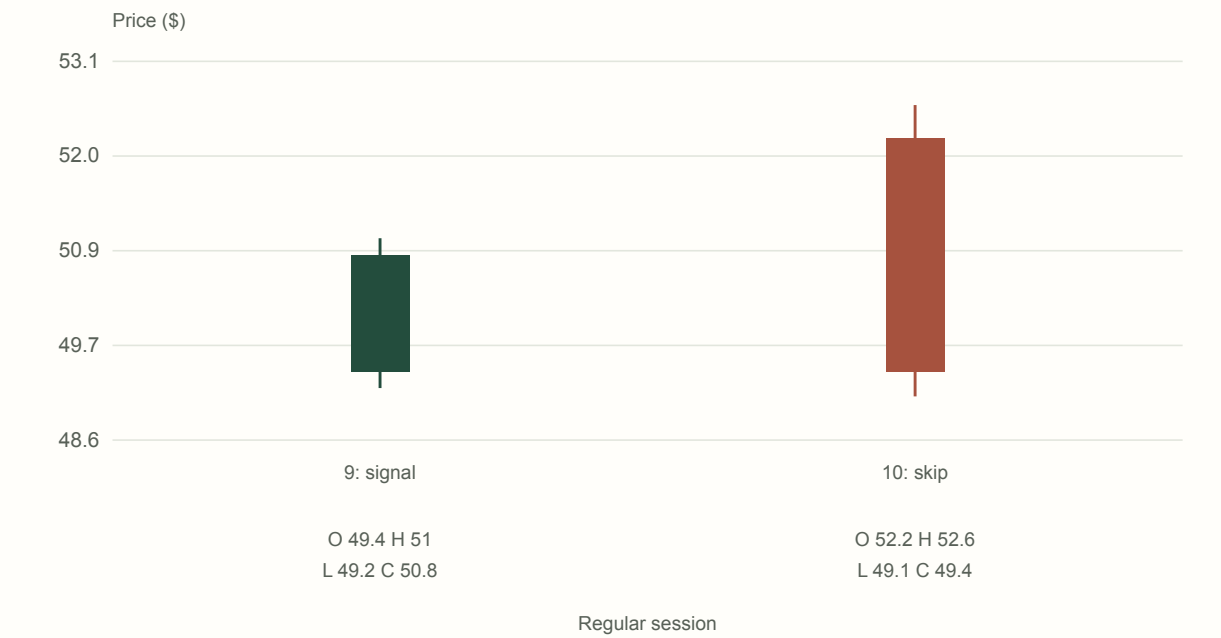
A closing breakout followed by a return below

Fictional session	Harbor closes	Predefined boundary	Signal threshold (strictly above)
1	48	50	50.5
2	49	50	50.5
3	49.8	50	50.5
4	49.2	50	50.5
5	48.7	50	50.5
6	49.6	50	50.5
7	49.9	50	50.5
8	49.4	50	50.5
9	50.8	50	50.5

Fictional session	Harbor closes	Predefined boundary	Signal threshold (strictly above)
10	49.4	50	50.5
11	48.8	50	50.5
12	49.2	50	50.5
13	48.5	50	50.5
14	49	50	50.5

FICTIONAL TEACHING ILLUSTRATION

The signal close and the ineligible next open



Same main Harbor example. Session 9 closes at $50.80; session 10 opens at $52.20, so the opening-only plan skips before the closing failure is known. OHLC values are fictional and do not supply transaction sequence.

▼ View the illustration's data

The signal close and the ineligible next open

Interval	Open	High	Low	Close
9: signal	49.4	51	49.2	50.8

Interval	Open	High	Low	Close
10: skip	52.2	52.6	49.1	49.4

Read a retest with the future covered

Use a separate retest example to practice chronological interpretation. At the signal close of $51, the fixed boundary is $50 and the visit area is $49.80–$50.20. Require a later low inside that area and a close above $50.50 in the same session, within three sessions after the signal. On the first later session, L50.30/C50.80 fails the visit condition. On the second, L50.10/C50.60 meets both conditions; the event becomes known only at that second close. A third session cannot be used to recognize it earlier.

Contrast two branches at that same second-session cutoff. L50.10/C50.40 visits but fails the closing condition. L49.00/C50.60 fails this specific low-inside-area rule because the measured low is outside it. The low and close do not prove that a trade occurred inside the area: eligible transaction prices can jump over intervening values. A different rule based on any traded touch needs the relevant transaction record or an explicit inference convention. State the field you measured. These alternatives show why a rule must be read literally before its usefulness is judged.

For a concrete counterexample, take two complete fictional eligible-price tapes. Tape A is 49.30 → 49.00 → 50.60. Tape B is 49.30 → 49.00 → 50.00 → 50.60. Both have O49.30/H50.60/L49.00/C50.60; both fail the low-inside-$49.80–$50.20 rule. Only B contains a traded touch at $50.00. A line drawn between A's $49.00 and $50.60 marks passes across the area on paper, but that connecting stroke does not supply a transaction. Without the tapes, their identical OHLC cannot distinguish the touch outcomes.

Identical OHLC does not establish an intermediate traded touch

Complete fictional tape	Ordered eligible prices ($)	Open ($)	High ($)	Low ($)	Close ($)
A	49.30 → 49.00 → 50.60	49.3	50.6	49	50.6
B	49.30 → 49.00 → 50.00 → 50.60	49.3	50.6	49	50.6

A WORKED EXAMPLE

A correct signal can end in a skip

At the session-9 close, $50.80 exceeds the frozen $50.50 threshold. At the session-10 open, $52.20 exceeds the entry ceiling, so the paper plan skips. At the session-10 close, $49.40 is below the $50 boundary and meets the failure definition. The three statements are compatible. No position was opened, so neither a hypothetical stop loss nor the foregone target belongs in realized P&L.

Pause & try it

Apply the supplied main Harbor rule at the session-9 close, session-10 open, and session-10 close. State the signal, entry decision, failure classification, and realized trade P&L at each cutoff. For the target comparison branch, calculate entry notional and cash debit including entry cost, planned price risk, cost-inclusive ordinary loss, and net target profit in dollars and R.

At the session-9 close, 50.80 > 50.50 establishes a signal; the next open and later failure are still unknown. At the session-10 open, 52.20 > 50.80 requires a skip. At its close, 49.40 < 50.00 establishes failure inside the three-session window. There is no trade P&L, because there was no entry; a skip is a retained opportunity record rather than a zero-R executed trade.

For the explicitly modeled $50.80 target branch: entry notional = 50 × $50.80 = $2,540; entry cash debit = $2,542.50 including the $2.50 entry charge. Planned price risk = 50 × $2 = $100; ordinary loss including $5 expense = $105. Target price profit = 50 × $4 = $200; net profit = $195; net R = 195/100 = +1.95R. The denominator is frozen filled-price risk, not notional or the $105 cost-inclusive loss.

The comparison branches are invented continuations for inspecting the same plan. Their count is not a sample of strategy success rates. A bar containing both exit prices would require the declared ambiguity policy.

GUIDED PRACTICE

Follow the retest clock

Using the retest rule, classify day +1 L50.30/C50.80 and day +2 L50.10/C50.60. What changes if day +2 closes at $50.40?

Day +1 misses the low-inside-$49.80–$50.20 condition. Day +2 meets the visit and close-above-$50.50 conditions, recognized at its close. With C50.40, the visit occurs but the closing condition fails; neither version licenses a day +1 recognition.

INDEPENDENT PRACTICE

A gap changes the realized loss

A comparison branch first models 50 shares bought at $50.80. The following session opens at $47.80, below the $48.80 stop. Under the supplied gap rule and $5 expense, calculate the exit, net P&L, and R. Explain why the ordinary $105 modeled loss is not the answer.

The gap rule exits at the assumed $47.80 open. Price loss = 50 × ($50.80 − $47.80) = $150; net P&L = −$155; R = −155/100 = −1.55R. The trigger was skipped by the gap, so an exact $48.80 fill cannot be assumed. The $105 figure described an ordinary exact-trigger exit, not a maximum possible loss.

INDEPENDENT PRACTICE

Keep R tied to the eligible fill

A separate permitted branch opens at $50.40 and models the same 50-share entry, $48.80 stop, $54.80 target and $5 round-trip charge. Calculate frozen 1R, entry cash debit, net target result and net ordinary-stop result in dollars and R. How much of the maximum-entry cash reserve is released?

The $50.40 opening entry is above $48.80 and at or below $50.80, so it is eligible. Frozen 1R = 50 × ($50.40 − $48.80) = $80. Entry notional is $2,520; cash debit is $2,522.50. The $2,542.50 maximum-entry reserve releases $20.

Target gross profit = 50 × $4.40 = $220; net = $215; net R = 215/80 = +2.6875R. Ordinary-stop net loss = −$80 − $5 = −$85, or −1.0625R. Reusing the maximum-entry $100 denominator would conceal the smaller filled-price risk.

INDEPENDENT PRACTICE

A traded touch needs a traded price

A different retest area is $72.80–$73.20 inclusive. For each complete fictional eligible-price tape below, reconstruct OHLC and apply two rules: the completed low must be inside the area; or at least one eligible trade must be inside it. Both rules additionally require a close strictly above $73.50. Explain what an OHLC-only observer cannot distinguish.

Complete unfamiliar eligible-price tapes (fictional USD)

Tape	All eligible trade prices, in order ($)
C	72.30 → 72.00 → 73.60
D	72.30 → 72.00 → 73.00 → 73.60

Both summarize as O72.30/H73.60/L72.00/C73.60. The low is below the area, so neither passes the low-inside-area rule; both satisfy the separate close-above-$73.50 condition.

C has no eligible trade in $72.80–$73.20, while D contains $73.00, so only D passes the traded-touch rule. The jump from $72.00 to $73.60 does not establish intermediate transactions. With only the identical OHLC, traded touch is unresolved; it cannot be inferred by drawing a continuous line.

CHECK YOUR UNDERSTANDING

One question before you go

At a $50.80 qualifying close, the plan permits only a next open at or below $50.80. The next open is $52.20 and the close is $49.40. Which record follows the supplied rule?

A. Signal at the earlier close; buy at $50.80 because the later low passed it; stop loss.

B. Signal at the earlier close; skip at the open; chart failure at the later close.

C. No signal at the earlier close because the next close was below the boundary.

Answer: B. The opening-only policy skips, while signal recognition and later closing failure remain distinct observations.

Sources & further reading

- StockCharts ChartSchool · Support and resistance ↗
  Breaks of previously defined trading areas, approximate zones, role-change and failed-crossing vocabulary. The numerical breakout and failure rules in this book are original fictional examples. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- StockCharts ChartSchool · Symmetrical triangle ↗
  Converging lower highs and higher lows; distinction between pattern geometry and observed breakout direction; filter and retest conventions. No performance percentages are adopted here. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.
- SEC / Investor.gov · Types of orders ↗
  Market, limit, and stop order distinctions; execution prices and fills are not guaranteed by a chart level. Original access recorded 2026-10-01. Latest access check 2026-10-12: web reader accessible.
- Fidelity · Why to consider stock volume ↗
  Share-volume definition and comparison with historical activity. Original access recorded 2026-10-01. Latest access check 2026-10-12: retrieved.

Teaching examples are fictional unless labelled historical. Inspect this chapter's evidence. Sources support the specified concepts; they do not validate a profitable trading strategy.

Support, resistance & price areas ↗ Classical chart patterns ↗ From chart idea to trade plan ↗

Evidence before conviction ↗

Practice & review this part →